

Knowing everyone's truthfulness and one's subordinates' codes

An affirmative answer to Question 3-1-1 of S. A. Alexander, *The Theory of Several Knowing Machines* (2013)

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Status.

- **Kind of proof:** a hand proof. Not machine-checked.
- **Checks:** it was written by one AI system and checked step by step by another; see the repository's provenance file. No independent human review.
- **Novelty:** the argument combines two techniques already in Alexander's work. What is new is their combination, and the answer to a question the dissertation lists as open.
- **Priority:** targeted searches found no earlier resolution. That is not a full priority check.

Source. S. A. Alexander, *The Theory of Several Knowing Machines*, doctoral dissertation, Ohio State University, 2013 (OhioLINK ETD osu1366362999). The question is from §1.1, Definition 1.1.1 and Figure 1. Also cited: S. A. Alexander, *Self-referential theories*, J. Symbolic Logic 85(4), 2020, [arXiv:2008.11535](https://arxiv.org/abs/2008.11535).

The question. Let R be an r.e. well-founded relation on $\mathbb{N}$, where $j R i$ is read "agent j is subordinate to agent i ". Question 3-1-1 asks whether it is consistent with self-doubting Epistemic Arithmetic that every agent m satisfies three conditions:

- **(3-x-x)** m knows the truthfulness of everyone: $K_m(\text{ucl}(K_n\varphi \rightarrow \varphi))$ for every n and every φ ;
- **(x-1-x)** m knows codes for its subordinates: $\exists e K_m \forall x (K_n\varphi \leftrightarrow x \in W_e)$ for every $n R m$ and every φ with a single free variable x ;
- **(x-x-1)** m knows the mechanicalness of its subordinates: $K_m \exists e \forall x (K_n\varphi \leftrightarrow x \in W_e)$ for $n R m$.

The dissertation answers 24 of the 27 questions in its matrix. It leaves the three Questions 3-1-x open and instead answers a variant, (3-1-2)', in which m 's knowledge of the truthfulness of non-subordinates is restricted to formulas that do not expose K_m .

Answer. Yes, for every r.e. well-founded R . R need not be transitive.

1. Setting (the dissertation's §1.2)

The language L_ω is the language of Peano arithmetic, extended by modal operators K_i ($i \in \mathbb{N}$).

The base logic (Definition 1.2.3).

- A structure interprets $K\varphi[s]$ by an arbitrary truth value, subject to three conditions:
 - independence from variables that are not free;
 - invariance under alphabetic variants;
 - weak substitution of *variables*.

- Validity means truth in all structures.
- Semantic consequence $\models$ is effective (Theorem 1.2.6), and “consistent” means satisfiable.
- φ^s substitutes numerals for free variables, including inside operators, so $(K\varphi)^s \equiv K(\varphi^s)$ (Lemma 1.2.12). Numeral instances of valid formulas need not be valid (Example 1.2.13); this is why *Assigned Validity* (every φ^s with φ valid) is used below.

Knowing machines.

- A family $K_i \mapsto T_i$ of r.e. theories determines the **intended structure** M_T : standard arithmetic, with $M_T \models K_i\varphi[s]$ iff $T_i \models \varphi^s$.
- The family is a *family of knowing machines satisfying self-doubting EA* if M_T satisfies the following:
 - for each i , the i -Validity, i -Reasoning, i -Introspection and i -Truth axioms;
 - $K_i\psi$ for every instance ψ of the first three;
 - Peano arithmetic for L_ω (induction for every formula);
 - K_i applied to each PA axiom.

2. Construction

Fix effective codings. For each i and each parameter q , let $G_i(q)$ be the following set of sentences.

1. The PA axioms of L_ω , and Assigned Validity.
2. i -Validity, i -Reasoning and i -Introspection (universal closures).
3. **Full reflection:** $\text{ucl}(K_j\varphi \rightarrow \varphi)$ for **every** j and every φ , including $j = i$ and formulas exposing K_i .
4. **Codes:** for $j \text{ R } i$ and φ with $FV(\varphi) = \{x\}$,

$$\forall x (K_j\varphi \leftrightarrow x \in W_{s(q,j,\varphi)}), \quad W_{s(q,j,\varphi)} = \{m : \langle \ulcorner \varphi \urcorner, j, m \rangle \in W_q\},$$

where s is an s-m-n function and the index is a numeral.

Let $A_i(q) = \{K_i^r\sigma : \sigma \in G_i(q), r \geq 0\}$, and let $T_i(q)$ be its set of consequences. Because R is r.e., this family is uniformly r.e. in q and i . Take a total computable f with $W_{f(q)} = \{\langle \ulcorner \varphi \urcorner, j, m \rangle : FV(\varphi) = \{x\}, T_j(q) \models \varphi(x|\vec{m})\}$. The recursion theorem then gives q with $W_q = W_{f(q)}$. Fix this q and write T_i for $T_i(q)$. Then

$$m \in W_{s(q,j,\varphi)} \iff T_j \models \varphi(x|\vec{m}). \quad (1)$$

Lemma 1 (closure). For every sentence θ , if $T_i \models \theta$ then $T_i \models K_i\theta$.

Proof. Finitely many $\sigma_l \in A_i$ make $\sigma_1 \rightarrow \dots \rightarrow \sigma_r \rightarrow \theta$ valid.

- K_i of that implication is an i -Validity instance.
- Each $K_i\sigma_l$ belongs to A_i .
- r applications of i -Reasoning give $K_i\theta$.

All the formulas involved are sentences. $\square$

3. Proof that the family works

The provisional structure. Define M on standard arithmetic, **by recursion on formula complexity**:

$$M \models K_i\varphi[s] \iff T_i \models \varphi^s \text{ and } M \models \varphi[s]. \quad (\text{F})$$

This is the truth-filtered interpretation of the dissertation's dual proof (§2.3). The recursive call is on a proper subformula, and the consequence relation on the left is fixed syntax, so (F) presupposes no soundness.

M is a structure:

- **Free variables:** clear.
- **Alphabetic variants:** they are logically equivalent, and the truth clauses agree by induction.
- **Weak substitution:** $(\varphi(x|y))^s$ and $\varphi^{s(x|s(y))}$ are the same sentence.

Induction on formulas, using Lemma 1.2.12 at K, gives numeral-instance invariance: $M \models \varphi[s] \Leftrightarrow M \models \varphi^s$.

Step 1: every generator in items 1-3 is true in M, with no soundness assumption.

- **PA:** holds because arithmetic is standard; modal induction follows from metatheoretic induction applied to each defined set.
- **Assigned Validity:** validity plus numeral invariance.
- **i-Validity:** $T_i \models \varphi^s$ by Assigned Validity, and $M \models \varphi[s]$ by validity.
- **i-Reasoning:** modus ponens holds on both sides of (F).
- **i-Introspection:** Lemma 1 gives $T_i \models K_i(\varphi^s) \equiv (K_i\varphi)^s$, and the truth half of (F) is the hypothesis.
- **Reflection:** for every j it is exactly the truth half of (F).

Step 2: $M \models T_i$ for every i, by well-founded induction on R. Assume $M \models T_j$ for all j R i.

- **Codes.** For such j, the truth half of (F) is redundant. So for unary φ , $M \models K_j\varphi[x \mapsto m] \Leftrightarrow T_j \models \varphi(x|\tilde{m}) \Leftrightarrow m \in W_{s(q,j,\varphi)}$ by (1). The code generators are true.
- **The prefixes $K_i^r\sigma$.** They are true by induction on r: $\sigma \in T_i$, and σ is true.
- **Conclusion.** Every member of A_i is true, hence so is every consequence: $M \models T_i$.

The induction is sound because M is fixed before it starts. What is proved by induction is the *statement* " $M \models T_i$ ". It is not a fixed-point definition.

Step 3: removing the filter. Every T_i is now true in M, so the truth half of (F) is redundant everywhere and $M = M_T$. The requirements are then met:

- **Self-doubting EA:** Step 1 shows the axioms of knowledge hold. T_i contains PA, the known versions (by closure), and truth.
- **(3-x-x)** is item 3.
- **(x-1-x)** is item 4, which also gives (x-x-1).

Theorem. For every r.e. well-founded R there is a uniformly r.e. family of knowing machines satisfying self-doubting Epistemic Arithmetic and 3-1-1. In it, each machine also knows its own truthfulness. ■

4. Remarks

- **Why this is easy, and 3-1-2 is not.** Adding self-mechanicalness, $\exists e\forall x(K_i\varphi \leftrightarrow x \in W_e)$, as a generator of T_i breaks Step 2. Under (F), its truth needs the intersection of an r.e. consequence set with a truth set to be r.e. Well-founded induction does not supply this, because the dependence is on T_i itself. Question 3-1-2 is treated in note 15 of this repository with ordinal stratification.
- **The comparison with Self-referential theories, Theorem 24.** That theorem uses the unfiltered structure M_T with the same R-induction, and obtains truthfulness only toward strict subordinates. Filtering makes reflection toward *everyone* free of charge. The price is paid only by the code axioms, and there well-foundedness suffices.

- **A necessary asymmetry.** Suppose truthful A knows B's truthfulness and a code e for B on $\delta(x) = x \notin W_x$. Put $g = \delta(e)$. Then A knows g and $\neg K_B g$, while B does not know g . Consequently:

- A's arithmetic theorems strictly extend B's;
- if B also knows A's truthfulness, then B cannot know that A knows g .

In the family above this asymmetry is realised, not contradicted.

- **Self-loops are genuinely excluded.** With $i R i$, the induction hypothesis in Step 2 would be its own conclusion. By Reinhardt's argument, a machine that knows its own truthfulness and its own code is impossible.